

A test can fail if the code generates a Python error or if an assertion fails.

Here's a new test in `test_area_functions.py` that checks that `None` is returned for negative inputs:

```
def test_negative_inputs(self):  
    """Test negative inputs."""  
    area = af.area_rect(-3, 4)  
    self.assertEqual(area, None)
```

This test fails; here's the output:

```
$ python test_area_functions.py  
F.  
=====  
FAIL: test_negative_inputs  
Test negative inputs for area_rect().  
-----  
Traceback (most recent call last):  
  File "test_area_functions.py", line 15:  
    self.assertEqual(area, None)  
AssertionError: -12 != None  
-----  
Ran 2 tests in 0.000s  
FAILED (failures=1)
```

A failing test shows an F: the output shows the failed test and prints its docstring, the failed assert statement, and the assert statement's values. A final summary shows the number of tests, their runtime, and the failures.